Haibin Guan

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RESEARCH INTERESTS

Metabolomics • Environmental Exposomics • Multi-omics Integration • High-dimensional Mediation Analysis • Statistical Methods for Omics Data

WORK EXPERIENCE

Icahn School of Medicine at Mount Sinai — New York, United States
Data Analyst I — 5/2022 – 5/2024
Data Analyst II — 5/2024 – Present

- Conduct metabolomics and exposomics analyses including data preprocessing, normalization, and statistical modeling on high-dimensional omics datasets.
- Collaborate with biostatistics and environmental health research teams on method development and advanced statistical analyses for epidemiologic studies.

INFOTECHSoft — Miami, FL, United States
Biomedical Data Analyst Intern — 1/2018 – 7/2018

- Contributed to the development of a metabolomics data integration system for multi-platform biological data.
- Collaborated with bioinformatics researchers and software engineers to support large-scale microbiome data integration and analysis.

EDUCATION

Florida International University — Miami, FL, United States
MS in Data Science — 1/2017 – 8/2018
Transferred to MS in Computer Science — 8/2018 – 5/2019

Wenzhou Medical University — Wenzhou, Zhejiang, China
Bachelor in Preventive Medicine — 9/2011 – 7/2016

SELECTED RESEARCH PROJECTS

Multi-omics Integration Analysis of Environmental Exposures and Cancer Risk — 2025 – Present
Icahn School of Medicine at Mount Sinai

- Developed a stability-driven multi-omics integration framework combining sparse generalized canonical correlation analysis (SGCCA) with repeated cross-validation and out-of-sample projection to evaluate component- and feature-level reproducibility.
- Applied the framework to untargeted metabolomics and Olink inflammation proteomics in a thyroid cancer case–control cohort ($n = 162$) to identify reproducible latent molecular signatures linked to thyroid cancer risk and PFOS exposure.
- First-author manuscript under revision at *Communications Biology* (preprint: *bioRxiv*, 2026); presented at MANA 2026 (poster) with an Early Career Members Travel Award.

Identification of Critical Windows of Susceptibility Using Time-varying Exposure Models — 2025 – Present
Icahn School of Medicine at Mount Sinai

- Applying statistical models for time-varying exposures/mixtures, including BDLIM, TDLMM or other tree-based distributed lag models (dlmtree), to identify critical windows of susceptibility.
- Investigating associations between longitudinal environmental exposures and maternal health outcomes while characterizing sensitive developmental periods.
- Manuscript in preparation.

Maternal PFAS Mixture Exposure and SARS-CoV-2 IgG: Metabolomic Mediation Analysis — 11/2023 – 04/2025
Icahn School of Medicine at Mount Sinai

- Conducted high-dimensional mediation analysis to examine whether maternal plasma metabolites mediate the inverse association between PFAS mixture exposure and SARS-CoV-2 IgG levels during pregnancy, using untargeted LC-HRMS metabolomics.

- Constructed PFAS mixture indices and positively and negatively associated metabolite indices using weighted quantile sum (WQS) regression, and evaluated their mediation effects individually and jointly, identifying a cumulative mediation effect of 83.8%.
- First-author publication in *Science of the Total Environment* (2025); presented as a plenary talk at MANA 2024.

Metabolomics Data Processing and Batch Correction Pipeline 02/2023 – 01/2024
Icahn School of Medicine at Mount Sinai

- Developed automated preprocessing workflows for untargeted metabolomics data using R-based tools (XCMS, IPO) for peak detection, retention time alignment, and feature grouping.
- Evaluated batch correction and missing-data imputation strategies and generated automated quality-control reports for LC/MS and GC/MS datasets.

PFAS Exposure and Thyroid Cancer Risk 10/2022 – 03/2023
Icahn School of Medicine at Mount Sinai

- Investigated associations between PFAS exposures and thyroid cancer risk using plasma metabolomics data from case-control pairs in the BioMe biobank.
- Applied regression-based epidemiologic models with covariate adjustment and inverse probability weighting to address potential selection bias.

DATA SCIENCE COMPETITIONS

TReNDS Neuroimaging Kaggle Competition 4/2020 – 6/2020
Participant Kaggle

- Built machine learning models to predict brain age and cognitive assessment scores using functional connectivity and structural MRI features.
- Applied PCA-based dimensionality reduction and trained regression models including SVR, KNN, and ElasticNet.
- Achieved a Silver Medal ranking (27 / 1047 teams).

Multiple Myeloma DREAM Challenge 7/2017 – 12/2017
Participant Florida International University

- Developed predictive models for gene expression datasets using feature selection and ensemble machine learning approaches.
- Applied SMOTE sampling, differential expression analysis (limma), and multiple classifiers including SVM, Random Forest, and Neural Networks.
- Achieved first place in the second round of the competition among 40 teams.

TECHNICAL SKILLS

Programming: R (primary), Python, MATLAB, Linux, LaTeX, HTML/CSS
Multi-omics Integration: SGCCA/RGCCA, DIABLO (mixOmics), MOFA/MOFA+, stability selection, resampling-based validation
Mediation & Mixtures: High-dimensional mediation (HIMA, bama), compositional mediation (CCMM), WQS, Qg-comp, BKMR
Time-varying Exposures: BDLIM, TDLM/TDLMM (dlmtree)
Other Statistical Methods: Regression and Cox models, inverse probability weighting, prediction-powered inference, Bayesian modeling (brms)
Metabolomics: XCMS, IPO, GNPS, MetaboAnalyst; batch correction (ComBat, EigenMS, WaveICA, TIGERr, QCRSC)
Omics Data Types: Untargeted LC/GC-MS metabolomics and exposomics, Olink proteomics, microRNA, mtDNA
Data Types: Untargeted LC/GC-MS metabolomics and exposomics, Olink proteomics, microRNA, mtDNA, gene expression (RNA-seq, microarray), MRI imaging features
Machine Learning: PyTorch, TensorFlow, scikit-learn
Cloud Computing: AWS

PUBLICATIONS

- **Guan H**, van Gerwen M, Kim-Schultz S, Colicino E, Dolios G, Petrick LM. Stability-driven multi-omics integration for reproducible latent structure. *bioRxiv*. 2026. doi:10.64898/2026.06.23.734064. (Under revision, *Communications Biology*)

- Aldana SI, Merced-Nieves F, Midya V, Yu X, Svensson K, Thompson JW, et al. The chemical exposome during peri-pregnancy, maternal sleep outcomes post-pregnancy, and the potential mediating role of body mass index. *Environ Pollut*. 2026;129220.
- Midya V, Maroli AS, Picker M, Dolios G, Valvi D, Nguyen I, Kim T, Rendon A, Chen R, Weinstein K, **Guan H**, Joseph G, Eggers J, Torres-Olascoaga LA, Ravisekar R, Srinath B, Ranchadiya R, Andra SS, Achaintre D, Gennings C, Téllez-Rojo MM, Wright RO, Arora M, Rosa MJ, Niedzwiecki M, Torres J, Alcala CS, Lane JM, Eggers S, Colombel JF, Peter I, Petrick LM, Agrawal M. Per- and polyfluoroalkyl substances (PFAS) in early life is associated with childhood intestinal inflammation: analyses of three birth cohorts. *Clin Gastroenterol Hepatol*. 2026.
- Petrick LM, Dassanayake PS, Gennings C, **Guan H**, Akintunde B, Martinez M, Regmi P, Hertz-Picciotto I, Arora M, Bennett DH. Fetal exposure to persistent organic pollutants and childhood autism risk. *Environ Res*. 2026;306:125179.
- Joseph G, Renard-Dausset E, Invernizzi A, Chen J, **Guan H**, Wright RJ, Wright R, Rosa MJ, Horton M, Téllez-Rojo MM, Torres Calapiz M, Colicino E, Vrijheid M, Foraster M, Bustamante Pineda M, Rivas I, Llurba Olivé E, Maitre L, Petrick LM. Association between per- and polyfluoroalkyl exposures and postpartum psychological symptoms: Evidence from two population-based pregnancy cohorts. *Environ Res*. 2026;305:124923.
- Svensson K, India-Aldana S, Saddiki H, Petrick LM, **Guan H**, Hernández-Chávez C, Téllez-Rojo MM, Wright RO, Wright RJ, Colicino E. Pregnancy stress exposures and postpartum serum metabolomic profiles in mothers. *Metabolites*. 2026;16(5):312.
- Joseph G, van Gerwen M, Kim-Schulze S, Chen Z, Nadkarni GN, **Guan H**, Petrick LM. Inflammatory proteins in pre-diagnosis versus at-diagnosis samples associated with differentiated thyroid cancer. *Int J Cancer*. 2026;158(8):2072–2082.
- Metayer C, Gennings C, Morimoto LM, Mistry S, McGlynn KA, Dolios G, **Guan H**, Gonzaga CR, Andra SS, Petrick LM. Phthalates measured at birth and risk of testicular cancer in adolescents and young adults. *JNCI Cancer Spectr*. 2026;10(2):pkag011.
- Svensson K, Sanchez-Guerra M, Midya V, **Guan H**, Niedzwiecki M, Dolios G, Osorio-Yáñez C, Del Razo LM, Petrick LM, et al. Postpartum metabolomics profile predicts 8-years postdelivery women's subclinical atherosclerosis measures. *Sci Rep*. 2025;15(1):34939.
- Morimoto LM, Metayer C, Dolios G, Wiemels JL, Ma X, **Guan H**, Maroli A, Petrick LM. Per- and polyfluoroalkyl substance concentrations during pregnancy and at birth and risk of childhood acute lymphoblastic leukemia. *Environ Res*. 2025;122436.
- **Guan H**, Chen J, Kaur K, Amreen B, Lesseur C, Dolios G, Andra SS, Narasimhan S, Pulivarthi D, Midya V, De Witte LD, Bergink V, Rommel AS, Petrick LM. High-dimensional mediation analysis to elucidate the role of metabolites in the association between PFAS exposure and reduced SARS-CoV-2 IgG in pregnancy. *Sci Total Environ*. 2025;980:179520.
- Petrick LM, **Guan H**, Page GP, Dolios G, Niedzwiecki MM, Wright RO, Wright RJ. Comparison of maternal venous blood metabolomics collected as dried blood spots, dried blood microsamplers, and plasma for integrative environmental health research. *Environ Int*. 2024;187:108663.
- Petrick LM, Niedzwiecki MM, Dolios G, **Guan H**, Tu P, Wright RO, Wright RJ. Effects of storage temperature and time on metabolite profiles measured in dried blood spots, dried blood microsamplers, and plasma. *Sci Total Environ*. 2024;912:169383.
- van Gerwen M, Colicino E, **Guan H**, Dolios G, Nadkarni GN, Vermeulen RCH, Wolff MS, Arora M, Genden EM, Petrick LM. Per- and polyfluoroalkyl substances (PFAS) exposure and thyroid cancer risk. *eBioMedicine*. 2023;97:104831.
- Soto PJ, Ilmer I, **Guan H**, Li J. Lightweight projective derivative codes for compressed asynchronous gradient descent. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. 2022:20444–20458.
- Ruiz-Perez D, **Guan H**, Madhivanan P, Mathee K, Narasimhan G. So you think you can PLS-DA? *BMC Bioinformatics*. 2020;21(Suppl 1):2.
- Zhang T, **Guan H**, Li F, He F. Modeling of influenza-like illness prediction based on Elman neural network. *Journal of Preventive Medicine*. 2019:113–118.
- Soto P, **Guan H**, Li J. Locally random P-adic alloy codes with channel coding theorems for distributed coded tensors. *arXiv*.
- Joseph G, DeBolt CA, Maroli A, **Guan H**, Ncube L, Gennings C, Petrick LM. Mid-pregnancy environmental chemical exposures are associated with maternal blood pressure in a prospective pilot study of Black women. *SSRN* (preprint). 2026.

PRESENTATIONS

- Stability-Driven Multi-Omics Integration for Reproducible Latent Structure. MANA 2026 Poster Presentation, Davis, CA.
- High-dimensional mediation analysis to elucidate the role of metabolites in the inverse association between PFAS exposure and SARS-CoV-2 IgG in pregnancy. MANA 2024 Plenary Presentation, Tampa, FL.

SOFTWARE AND APPLICATIONS

- Worklists Generator: R Shiny application designed to generate Orbitrap instrument worklists from SampleID files. https://haibin-guan.shinyapps.io/worklists_generator/
- BatCheck: R Shiny application for visualization and assessment of inter- and intra-batch effects in mass spectrometry data. https://haibin-guan.shinyapps.io/BatCheck/

PROFESSIONAL DEVELOPMENT

SHARP Multi-Omics Boot Camp — New York, NY, USA
Participant, Columbia University Multi-Omics Initiative — 07/2026

MANA WomiX Mentorship Program — USA
Participant — 4/2024 – 2/2025

Oxford Machine Learning Summer School — Oxford, United Kingdom
Participant — 7/2023

AWARDS & HONORS

- **Early Career Members (ECM) Travel Award**, Metabolomics Association of North America (MANA) 2026
- **SHARP Program Scholarship**, Columbia University 2026